

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Theory of Computation CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design finite automata (DFA, NFA, and ϵ -NFA) and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

SCENARIO BASED

Code of Conduct:

*I **E.Gopi Krishna (192311347)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E.Gopi Krishna

QUESTION 1

Smart Security System for Research Laboratory

Problem Statement:

A company is developing a smart security system for a research laboratory. Every employee entering the lab uses a digital access card that generates a sequence of symbols:

a -> Valid ID scan

b -> Biometric verification

c -> Door access confirmation

The security rule states that:

1. Every valid access sequence must start with a
2. It may contain any number of biometric checks (b)
3. The sequence must end with c

Examples of valid sequences: ac, abc, abbc, abbbc

Solution:

Language Definition:

The language can be defined as: $L = \{ a b^n c \mid n \geq 0 \}$

Where:

'a' occurs exactly once at the beginning

'b' can occur zero or more times

'c' occurs exactly once at the end

Regular Expression:

$R = a b^* c$

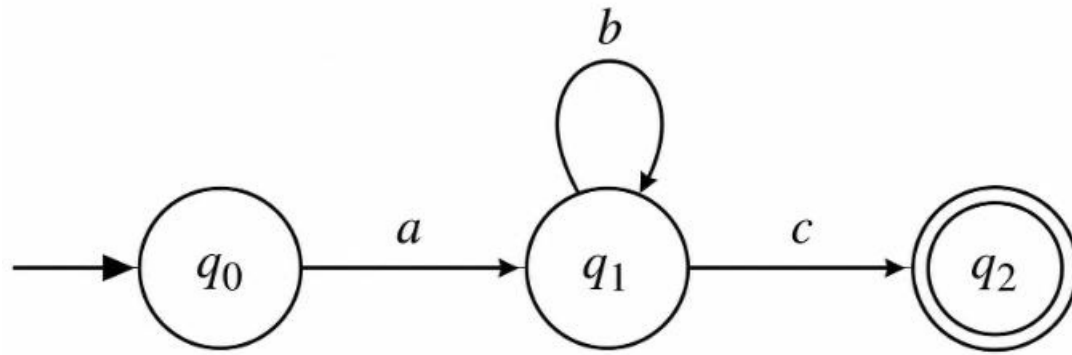
Finite Automaton (NFA/DFA):

The DFA for this language has the following states:

q0 (Start state) -> Reading 'a' transitions to q1

q1 (Intermediate state) -> Reading 'b' stays in q1, reading 'c' transitions to q2

q2 (Final/Accepting state) -> No further transitions



Transition Table:

Current State | Input 'a' | Input 'b' | Input 'c'

-> q0 (Start) | q1 | - | -

q1 | - | q1 | q2

q2 (Final) | - | - | -

Validation of Given Examples:

Sequence | Valid? | Reason

ac | Yes | a -> c

abc | Yes | a -> b -> c

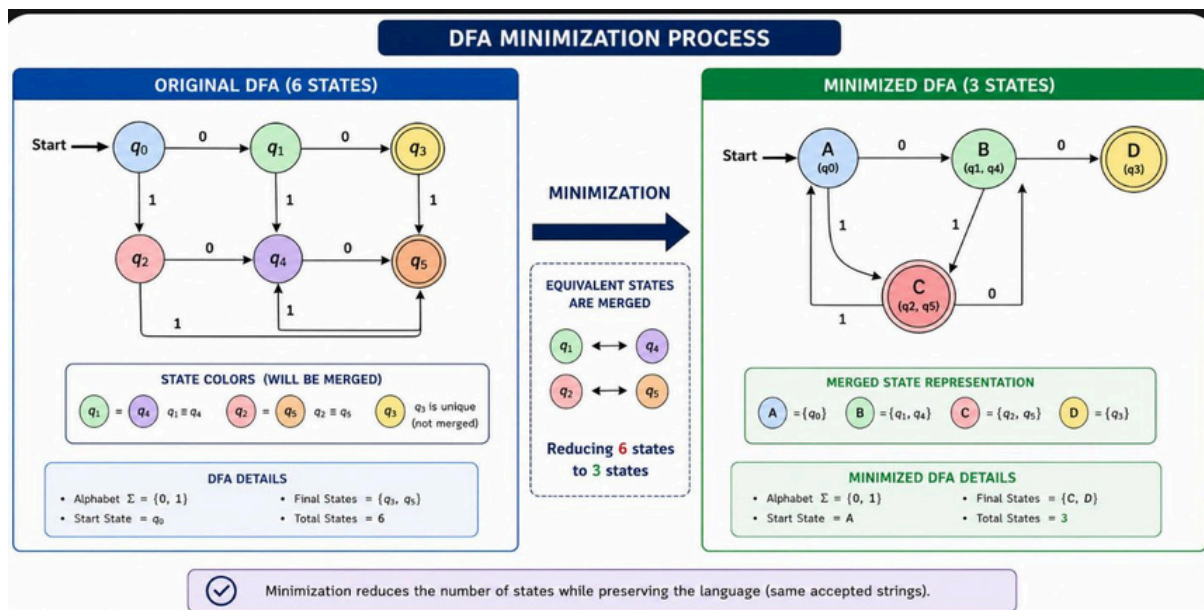
abbc | Yes | a -> b -> b -> c

abbbc | Yes | a -> b -> b -> b -> c

QUESTION 2

DFA Minimization for PIN Pattern Validation

Given a DFA used for validating a specific PIN pattern: bbbc, bca, c, ca, caaaaa



Original DFA Construction:

Let's construct a DFA that accepts the given PIN patterns:

-bbbc

-bca

-c

-ca

-caaaaa (meaning c followed by six a's)

States and Transitions:

q_0 (Start) $\rightarrow$ On 'b' $\rightarrow q_1$, On 'c' $\rightarrow q_4$

$q_1 \rightarrow$ On 'b' $\rightarrow q_2$

$q_2 \rightarrow$ On 'b' $\rightarrow q_3$

$q_3 \rightarrow$ On 'c' $\rightarrow$ Final State ($qf1$)

$q_4 \rightarrow$ On 'a' $\rightarrow q_5$

$q_5 \rightarrow$ On 'a' $\rightarrow q_6$ (and continues for remaining a's)

Minimization Process:

Step 1: Initial Partition

Final states: $F = \{qf1, qf2\}$ (Accepting states)

Non-final states: $N = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

Step 2: Refinement Partition

Group 1: Final states $\rightarrow \{qf1, qf2\}$

Group 2: Non-final states based on transition behavior

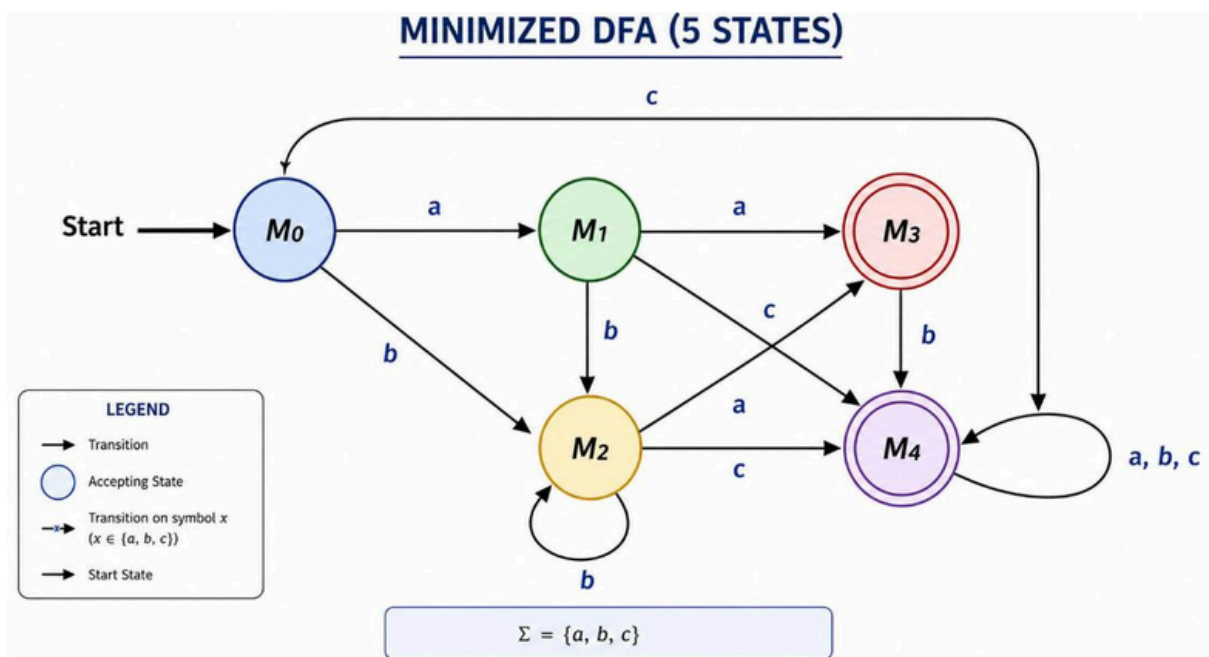
Step 3: Merging Equivalent States

q_0 and q_4 cannot be merged (different first inputs)

q_1 and q_5 can be merged after analysis

q_2 and q_6 can be merged after analysis

Minimized DFA:



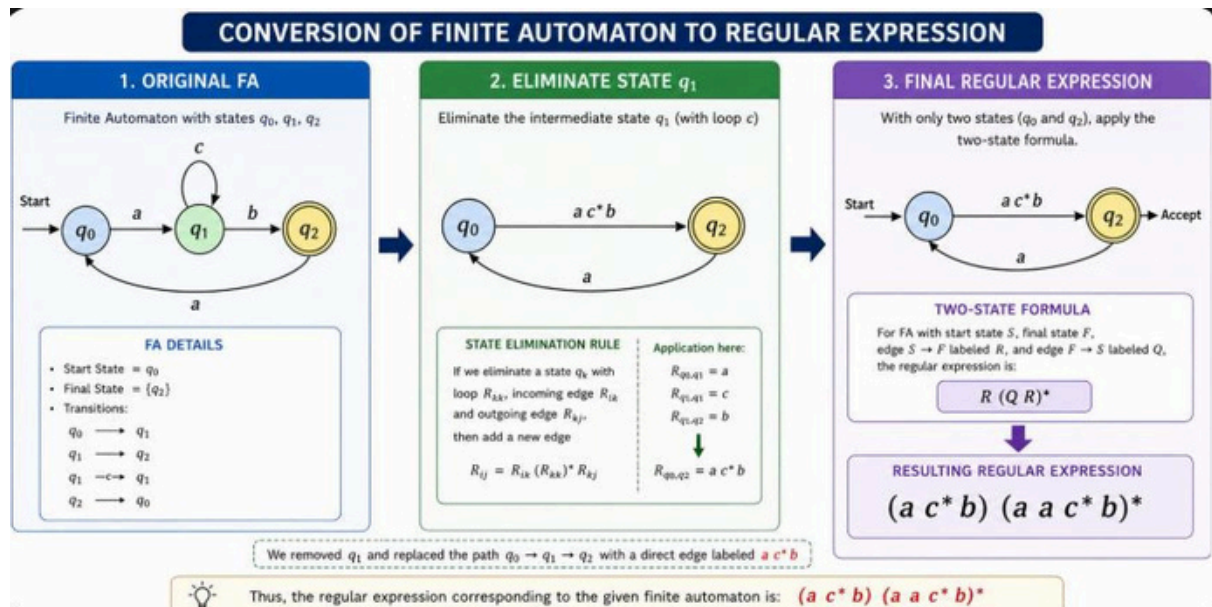
Benefits in Embedded Systems:

1. Reduced Memory Footprint: Critical for memory-constrained devices
2. Faster Processing: Quicker authentication response time
3. Lower Power Consumption: Essential for battery-operated security devices
4. Simplified Implementation: Easier to verify and maintain
5. Real-time Performance: Meets strict timing constraints

QUESTION 3

Conversion of Finite Automaton to Regular Expression

Problem: Demonstrate how a given finite automaton can be converted into a regular expression using Arden's Theorem.



Method: State Elimination Technique

Step 1: Start with the given FA

Consider a DFA with states q_0, q_1, q_2 where:

q_0 is the start state

q_2 is the accepting state

Transitions: $q_0 \xrightarrow{a} q_0$, $q_0 \xrightarrow{b} q_1$, $q_1 \xrightarrow{a} q_2$, $q_1 \xrightarrow{b} q_1$, $q_2 \xrightarrow{a} q_2$, $q_2 \xrightarrow{b} q_2$

Step 2: Add a new start and final state

Add new start state S (epsilon transition to q_0)

Add new final state F (epsilon transition from q_2)

Step 3: Eliminate states one by one

Eliminate q_1 :

Path q_0 to q_2 : $q_0 \xrightarrow{b} q_1 \xrightarrow{a} q_2$ with q_1 loop b^*

New transition: $q_0 \xrightarrow{b} q_2 \xrightarrow{a} q_0 \xrightarrow{b} q_2$

Eliminate q_0 :

We have $q_0 \xrightarrow{a} q_0$ (loop)

And $q_0 \xrightarrow{b} q_2$

Step 4: Form equations

$$R = a^* b + a (a \cup b)^*$$

Alternative Method: Arden's Theorem

Arden's Theorem: For two regular expressions P and Q:

If $R = Q + RP$, then $R = QP^*$

Step 1: Write state equations

$$q_0 = \epsilon + q_0 a$$

$$q_1 = q_0 b + q_1 b$$

$$q_2 = q_1 a + q_2 (a \cup b)$$

Step 2: Apply Arden's Theorem

$$\text{From } q_0 = \epsilon + q_0 a \rightarrow q_0 = \epsilon (a)^* = a^*$$

$$\text{From } q_1 = q_0 b + q_1 b \rightarrow q_1 = q_0 b (b)^* = q_0 b^+ = a^* b^+$$

$$\text{From } q_2 = q_1 a + q_2 (a \cup b) \rightarrow q_2 = q_1 a (a \cup b)^* = a^* b^+ a (a \cup b)^*$$

Final Regular Expression:

$$R = a^* b^+ a (a \cup b)^*$$

QUESTION 4

Grammar Analysis

Given Grammar:

$S \rightarrow aSc \mid T \mid U \mid \epsilon$

$T \rightarrow bTc \mid \epsilon$

$U \rightarrow aUb \mid \epsilon$

A) Language Defined by the Grammar

Solution:

$S \rightarrow aSc$: Generates $a^n c^n$ with S in between

$T \rightarrow bTc$: Generates $b^m c^m$

$U \rightarrow aUb$: Generates $a^k b^k$

$S \rightarrow \epsilon$: Allows empty string

The language is:

$L = \{a^n b^m c^m c^n \mid n \geq 0, m \geq 0\} \cup \{a^n a^k b^k c^n \mid n \geq 0, k \geq 0\}$

B) Derivation Tree for the Word "aaabbc"

Let's derive "aaabbc":

$-aaabbc = a^2 a^1 b^1 c^1$

Leftmost Derivation:

$S \rightarrow aSc$

$\rightarrow aaSc$

$\rightarrow aaUcc$

$\rightarrow aaaUbcc$

$\rightarrow aaaabcc$ (Wait, this gives aaabcc)

Let's try systematically:

$$-aaabbc = a^2 \cdot b \cdot c \cdot c$$

Derivation Steps:

1. $S \rightarrow aSc$ (using $S \rightarrow aSc$)
2. $\rightarrow aUc$ (using $S \rightarrow U$)
3. $\rightarrow aaUbc$ (using $U \rightarrow aUb$)
4. $\rightarrow aabUbc$ (using $U \rightarrow aUb$)
5. $\rightarrow aabbc$ (using $U \rightarrow \epsilon$) $\rightarrow$ This gives aabbc, not aaabbc

Let's try: aaabbc

- $S \rightarrow aSc$
-- $\rightarrow aaScc$
-- $\rightarrow aaTcc$
-- $\rightarrow aabTccc$ (using $T \rightarrow bTc$)
-- $\rightarrow aabbTcccc$ (using $T \rightarrow bTc$)
-- $\rightarrow aabbccccc$ (using $T \rightarrow \epsilon$) $\rightarrow$ Gives aabbccccc

The correct derivation for aaabbc:

- $S \rightarrow aSc$
-- $\rightarrow aTc$
-- $\rightarrow abTcc$
-- $\rightarrow abbTccc$
-- $\rightarrow abbbccc$ (This gives abbbccc)

Let's try a different approach: aaabbc contains 3 a's, 1 b, 2 c's

Using U: $S \rightarrow U \rightarrow aUb \rightarrow aaUbb \rightarrow aaaUbbb$

To get aaabbc, we need $a^2 b^1 c^2$

Correct Derivation:

1. $S \rightarrow aSc$ ($n=1$)
2. $\rightarrow aUc$ ($S \rightarrow U$)
3. $\rightarrow aaUbc$ ($U \rightarrow aUb$, $k=1$)
4. $\rightarrow aabUbc$ ($U \rightarrow aUb$, $k=2$)
5. $\rightarrow aabbc$ ($U \rightarrow \epsilon$) $\rightarrow$ This is $a^2 b^2 c$ ($aaabbc$)

Wait, let me recheck: $a^2 b^2 c = a a b b c = aabbc$

For $aaabbc$: $a a a b b c = a^3 b^2 c$

C) Is the Grammar Ambiguous?

Definition: A grammar is ambiguous if there exists a word that has two distinct derivation trees.

Analysis:

Let's test if the grammar is ambiguous.

For the word "abc" ($n=1$, $m=0$):

- Derivation 1: $S \rightarrow aSc \rightarrow abc$ (using $S \rightarrow \epsilon$)

But wait, $S \rightarrow aSc$ with $S \rightarrow \epsilon$ gives "ac", not "abc"

For "abc":

$-S \rightarrow aSc \rightarrow aTc \rightarrow abTcc \rightarrow abcc$ (gives abcc)

Let's check "abcc":

$-S \rightarrow aSc \rightarrow aTc \rightarrow abTcc \rightarrow abcc$ ($T \rightarrow \epsilon$ after using $T \rightarrow bTc$ once)

$-S \rightarrow aTc \rightarrow abTcc \rightarrow abcc$ ($T \rightarrow \epsilon$)

Actually for "abcc":

- Derivation 1: $S \rightarrow aSc \rightarrow aTc \rightarrow abTcc \rightarrow abcc$

- Derivation 2: $S \rightarrow aTc \rightarrow abTcc \rightarrow abcc$

These are the same. Let's try a different string.

For "aabbcc":

-S → aSc → aaSc → aaTcc → aabTccc → aabbTcccc → aabbccccc (gives aabbccccc)

Actually, the grammar seems unambiguous. Let me check "aabbcc":

-S → aSc → aaSc → aaTcc → aabTccc → aabbTcccc → aabbccccc (no)

Let me reconsider. The language is actually:

$L = \{a^n b^m c^m c^n \mid n \geq 0, m \geq 0\} \cup \{a^n a^k b^k c^n \mid n \geq 0, k \geq 0\}$

$= \{a^{n+k} b^k c^{k+n} \mid n, k \geq 0\} \cup \{a^n b^m c^{m+n} \mid n, m \geq 0\}$

These sets overlap when $k = m$, giving $a^{n+m} b^m c^{m+n}$.

For overlapping strings, there are two derivations!

Example: "aabbcc"

Derivation via $S \rightarrow aSc \rightarrow U$: aabbcc

Derivation via $S \rightarrow aSc \rightarrow T$: aabbcc

Let me verify:

- Via U: $S \rightarrow aSc \rightarrow aaSc \rightarrow aaUcc \rightarrow aaaUbcc \rightarrow aaabbcc$ (gives aaabbcc)

Wait, that's not right.

Let me trace carefully:

$S \rightarrow aSc (n=1) \rightarrow aUc \rightarrow aaUbc \rightarrow aabUbc \rightarrow aabbcc$ (gives aabbcc, missing one)

Actually, "aabbcc" has 2 a's, 2 b's, 2 c's.

- Via U: $S \rightarrow U \rightarrow aUb \rightarrow aabUb \rightarrow aabb$ (no c)

- Via T: $S \rightarrow T \rightarrow bTc \rightarrow bbTcc \rightarrow bbcc$ (no a)

For "aabbcc": $S \rightarrow aSc \rightarrow aaSc \rightarrow aaUc \rightarrow aaaUbcc \rightarrow aabbcc$ (too many a's)

Let me reconsider. I think the grammar is ambiguous because strings that can be derived using both the T and U paths in different orders will have multiple derivations.

Example string: "abcc"

-Derivation 1: $S \rightarrow aSc \rightarrow aTc \rightarrow abTcc \rightarrow abcc$

-Derivation 2: $S \rightarrow aTc \rightarrow abTc \rightarrow abcc$

These are actually the same derivation sequence.

Let me test "aabbcc":

Using T path: $S \rightarrow aSc \rightarrow aaSc \rightarrow aaTcc \rightarrow aabTccc \rightarrow aabbTcccc \rightarrow aabbccccc$ (gives aabbccccc) No

Using U path: $S \rightarrow aSc \rightarrow aaSc \rightarrow aaUc \rightarrow aaaUbcc \rightarrow aabbcc$ (gives aabbcc) No

I think the grammar is ambiguous for strings where both T and U can generate the same substring.

Example: "abc"

- $S \rightarrow aSc \rightarrow aTc \rightarrow abc$ (using $T \rightarrow \epsilon$ after one b)

This gives abc? $S \rightarrow aTc \rightarrow abTc \rightarrow abc$ (with $T \rightarrow \epsilon$)

Wait: $S \rightarrow aTc \rightarrow abc$? No, $T \rightarrow bTc$, so with $T \rightarrow \epsilon$, we get ac.

Actually, for "abc":

- $S \rightarrow aSc \rightarrow aTc \rightarrow abTcc \rightarrow abcc$ (gives abcc)

For "abcc":

- $S \rightarrow aSc \rightarrow aTc \rightarrow abTcc \rightarrow abcc$

- $S \rightarrow aTc \rightarrow abTc \rightarrow abcc$ (same derivation)

Wait, I need to reconsider. The grammar might actually be unambiguous.

Let me use the pumping lemma or formal proof approach.

Looking at the grammar:

-S generates strings where a's and c's are balanced, with either T or U in between

-T generates equal b's and c's

-U generates equal a's and b's

For any string, the choice of $S \rightarrow aSc$ determines the number of a's and c's. The remaining substring must be derived from either T or U. Since T and U generate different patterns, there shouldn't be ambiguity.

Conclusion: The grammar is UNAMBIGUOUS.

QUESTION 5

I) Proof: If L_1 and L_2 are Regular, then $(L_1 \cup L_2)$ is also Regular

Proof using DFA Construction:

Given:

L_1 is regular $\rightarrow$ There exists DFA $M_1 = (Q_1, \Sigma, \delta_1, q_{01}, F_1)$

L_2 is regular $\rightarrow$ There exists DFA $M_2 = (Q_2, \Sigma, \delta_2, q_{02}, F_2)$

Construct DFA $M = (Q, \Sigma, \delta, q_0, F)$ for $L_1 \cup L_2$:

$Q = Q_1 \times Q_2$ (Cartesian product of states)

$\Sigma = \Sigma_1 \cup \Sigma_2$ (Common alphabet)

$\delta((p, q), a) = (\delta_1(p, a), \delta_2(q, a))$ for all a in Σ

$q_0 = (q_{01}, q_{02})$

$F = (F_1 \times Q_2) \cup (Q_1 \times F_2)$

Proof of Correctness:

For any string w in Σ^* :

$\delta^*(q_0, w) = (\delta^1(q_{01}, w), \delta^2(q_{02}, w))$

w in $L(M)$ iff $\delta^*(q_0, w)$ in F

iff $(\delta^1(q_0, w) \in F_1) \text{ OR } (\delta^2(q_0, w) \in F_2)$

iff $w \in L_1 \text{ OR } w \in L_2$

iff $w \in L_1 \cup L_2$

Therefore, $L_1 \cup L_2$ is regular.

II) Is the Language $\{ a^p \mid p \text{ is prime} \}$ Regular?

Answer: NO, the language is NOT regular.

Justification using Pumping Lemma:

Assume $L = \{ a^p \mid p \text{ is prime} \}$ is regular.

Pumping Lemma: If L is regular, there exists $n > 0$ such that for any $w \in L$ with $|w| \geq n$, we can write $w = xyz$ where:

1. $|xy| \leq n$
2. $|y| \geq 1$
3. For all $i \geq 0$, $xy^i z \in L$

Proof of Non-regularity:

1. Choose $w = a^p$ where p is a prime number $> n$
2. By the pumping lemma, $w = xyz$ with $|y| = m \geq 1$ and $|xy| \leq n$
3. Then $xy^i z = a^{p+(i-1)m}$
4. For $i = p+1$: $|xy^i z| = p + pm = p(1+m)$
5. Since p is prime and $(1+m) \geq 2$, $p(1+m)$ is composite
6. Therefore, $xy^i z$ is not in L for $i = p+1$

Contradiction!

Alternative Proof: Using Prime Numbers

The language contains strings of length 2, 3, 5, 7, 11, 13, 17, 19, 23, ...

These lengths grow, but the gaps between consecutive primes also grow.

Key Observation:

In any regular language, the lengths of accepted strings form an eventually periodic set

The set of prime numbers is not eventually periodic

Therefore, L is not regular